09. Fractions: Concept & Equivalence

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Finding a unit fraction of an amount:

What is 1/5 of 30?

Answer: Divide by the denominator: 30 ÷ 5 = **6**.

Example 2 — Finding a non-unit fraction of an amount:

What is 2/7 of 35?

Answer: First find one seventh: 35 ÷ 7 = 5. Then multiply by 2: 5 × 2 = **10**.

Example 3 — Adding fractions with the same denominator:

Add 3/10 + 4/10, then simplify if possible.

Answer: Add the numerators, keep the denominator the same: 3/10 + 4/10 = **7/10**. This is already in simplest form (7 and 10 share no common factor other than 1).

Example 4 — Simplifying a fraction:

Simplify 6/9 to its lowest terms.

Answer: 6 and 9 share a common factor of 3: 6 ÷ 3 = 2, 9 ÷ 3 = 3. Simplified: **2/3**.

Example 5 — Multi-step word problem:

Jack has 18 sweets. He eats 1/3 of them, then his sister gives him 4 more. How many sweets does he have now?

Answer: First find a third: 18 ÷ 3 = 6 eaten. Remaining: 18 − 6 = 12. Then add the extra sweets: 12 + 4 = **16 sweets**.

Foundation

15 questions

1. In the fraction 2/5, which number is the numerator and which is the denominator?

2. What is 1/2 of 6? A) 2 B) 3 C) 4 D) 6

3. Shade 1/3 of a bar with 3 equal parts. What fraction is NOT shaded?

4. Write an equivalent fraction to 1/4 with a denominator of 12.

5. Which of these is larger? A) 2/9 B) 5/9 C) They are equal D) Cannot tell

6. In the fraction 4/7, which number is the numerator and which is the denominator?

7. What is 1/3 of 9?

8. A bar has 8 equal parts and 3/8 are shaded. Which fraction is NOT shaded? A) 3/8 B) 5/8 C) 8/8 D) 0/8

9. Write an equivalent fraction to 1/4 with a denominator of 8.

10. Which is larger: 2/9 or 5/9?

11. In the fraction 1/6, what is the denominator? A) 1 B) 6 C) 7 D) 16

12. What is 1/3 of 12?

13. Shade 5/6 of a bar with 6 equal parts. What fraction is NOT shaded?

14. Select ALL the fractions below that are equivalent to 1/2: A) 2/4 B) 3/6 C) 4/9 D) 5/10 E) 3/5

15. Which is larger: 4/7 or 5/7?

Standard

20 questions

1. 9/10 − 4/10

2. What is 3/5 of 25? A) 10 B) 15 C) 20 D) 25

3. Simplify 4/10 to its lowest terms.

4. Convert the improper fraction 11/4 to a mixed number.

5. 5/9 + 2/9 = ? A) 7/9 B) 7/18 C) 3/9 D) 1

6. 7/9 − 4/9

7. What is 3/7 of 14?

8. Which fraction is 4/8 simplified to its lowest terms? A) 1/4 B) 1/2 C) 2/3 D) 4/8

9. Convert the improper fraction 7/2 to a mixed number.

10. 2/7 + 3/7

11. 5/8 − 2/8 = ? Give your answer in its simplest form. A) 3/8 B) 1/4 C) 3/16 D) 5/16

12. What is 5/9 of 36?

13. Simplify 6/10 to its lowest terms.

14. Select ALL the fractions below that are already in their lowest terms: A) 4/10 B) 5/9 C) 6/9 D) 7/11 E) 8/9

15. 2/9 + 7/9

16. Jack has 10 stickers. 1/5 are given away, and the rest are shared equally between 2 friends. How many stickers does each friend get?

17. Convert 13/5 to a mixed number, then add 1 whole. A) 3 3/5 B) 2 3/5 C) 3 2/5 D) 2 2/5

18. Tom has 24 sweets and eats 1/3 of them. Then Mia gives him 5 more. How many does he have now?

19. 3/8 + 2/8, then simplify your answer if possible.

20. Which is bigger: 1/4 or 1/8? Convert 1/4 to a fraction with denominator 8 first. A) 1/4 B) 1/8 C) They are equal D) Cannot tell

Hard

5 questions

1. Mia has 16 stickers. 1/4 are given away, and the rest are shared equally between 3 friends. How many stickers does each friend get?

2. Convert 17/5 to a mixed number, then subtract 1/5. A) 3 1/5 B) 3 2/5 C) 2 4/5 D) 3 3/5

3. Priya has 28 sweets and eats 1/4 of them. Then Sam gives her 6 more. How many does she have now?

4. 3/8 + 4/8 = 7/8. Select ALL the fractions below that are equivalent to 7/8: A) 14/16 B) 21/24 C) 7/9 D) 35/40

5. Which is bigger: 2/5 or 3/8? Convert both to a common denominator of 40 first. A) 2/5 B) 3/8 C) They are equal D) Cannot tell

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